

Solution Requirements

Date	15 March 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	User Authentication	Users can securely register and log in using username and password before accessing the HDI Classification System.	High
2	Input Data Submission	Registered users can enter HDI indicators (Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per capita) to get the prediction.	High
3	Data Validation	The system validates all input values to ensure they are within the acceptable range. Missing or incorrect values are prompted with appropriate error messages.	High
4	HDI Prediction	The system processes validated data using a Random Forest Classification model to predict the HDI category (Low, Medium, High, Very High).	High

Page 1 of 2

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
5	Result Display	The system displays the predicted HDI category along with the score range, interpretation, and improvement suggestions on the result page.	High
6	Visualization & Insights	The system provides charts and graphs (such as country-wise HDI distribution) to help users analyze development trends and gain insights.	Medium
7	Dataset Management (Admin)	Administrators can upload/manage datasets, view prediction logs, and monitor system usage.	Medium
8	Export Reports	Users can export the prediction results and graphs as PDF or CSV for future reference.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should process user input and generate HDI prediction quickly.	Prediction generated within 2 seconds
2	Scalability	The application should support multiple users and handle large datasets without significant performance degradation.	Supports 500+ users and large datasets
3	Security & Data Privacy	User data should be protected using secure authentication, encrypted storage, and safe session management.	Password hashing, HTTPS , secure session management
4	Reliability & Availability	The system should be available whenever users access it and recover from minor failures.	99% uptime during deployment
5	Usability & Accessibility	The application should have a simple, responsive, and user-friendly interface accessible from desktop and mobile devices.	Responsive UI with easy navigation
6	Other (Maintainability)	The code should be modular, easy to maintain, and deployable on different operating systems.	Compatible with Windows, Linux , and cloud deployment

Page 2 of 2